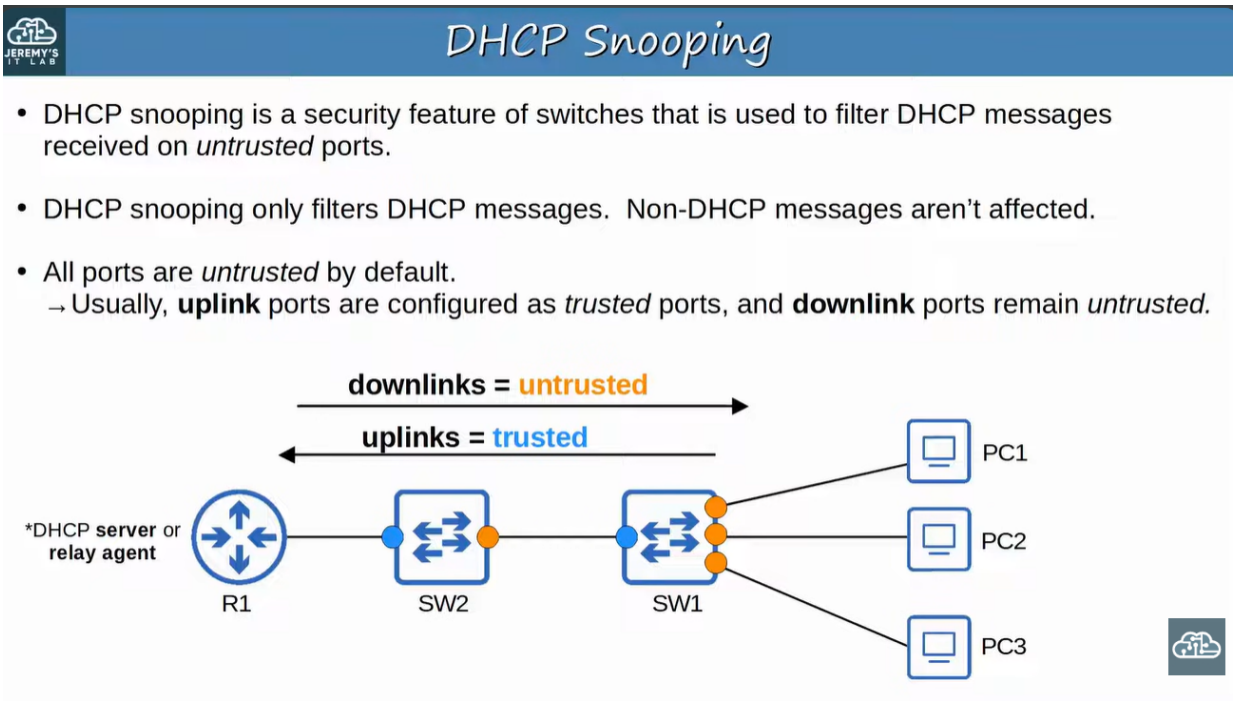
DHCP Snooping — Exam-Focused Notes

Goal: Prevent rogue DHCP servers and reduce DHCP-based attacks on a Layer-2 switched network.

1. DHCP Snooping — Core Idea

- DHCP Snooping is a **switch security feature** that filters DHCP messages received on **untrusted ports**.
- It filters **DHCP messages only**; normal non-DHCP traffic is not affected.
- All switch ports are **untrusted by default**.
- **Uplink** ports toward the legitimate DHCP server/relay are normally configured as **trusted**. Client-facing **downlink** ports normally remain **untrusted**.

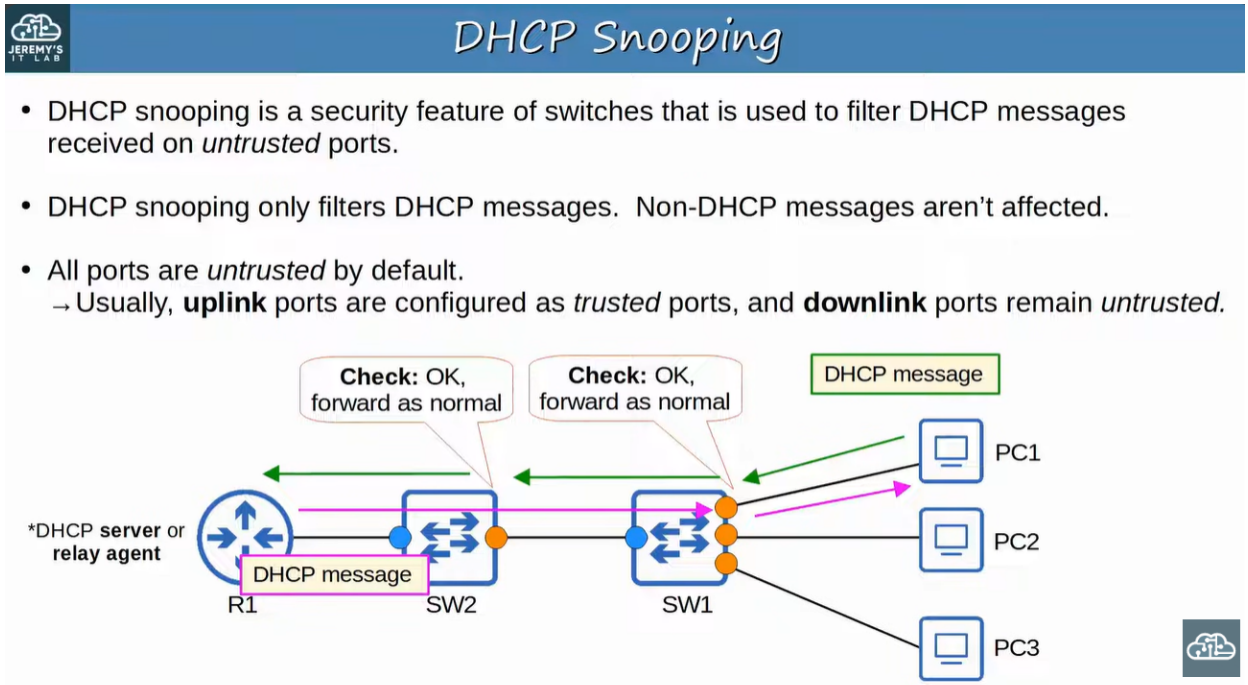
Image section:

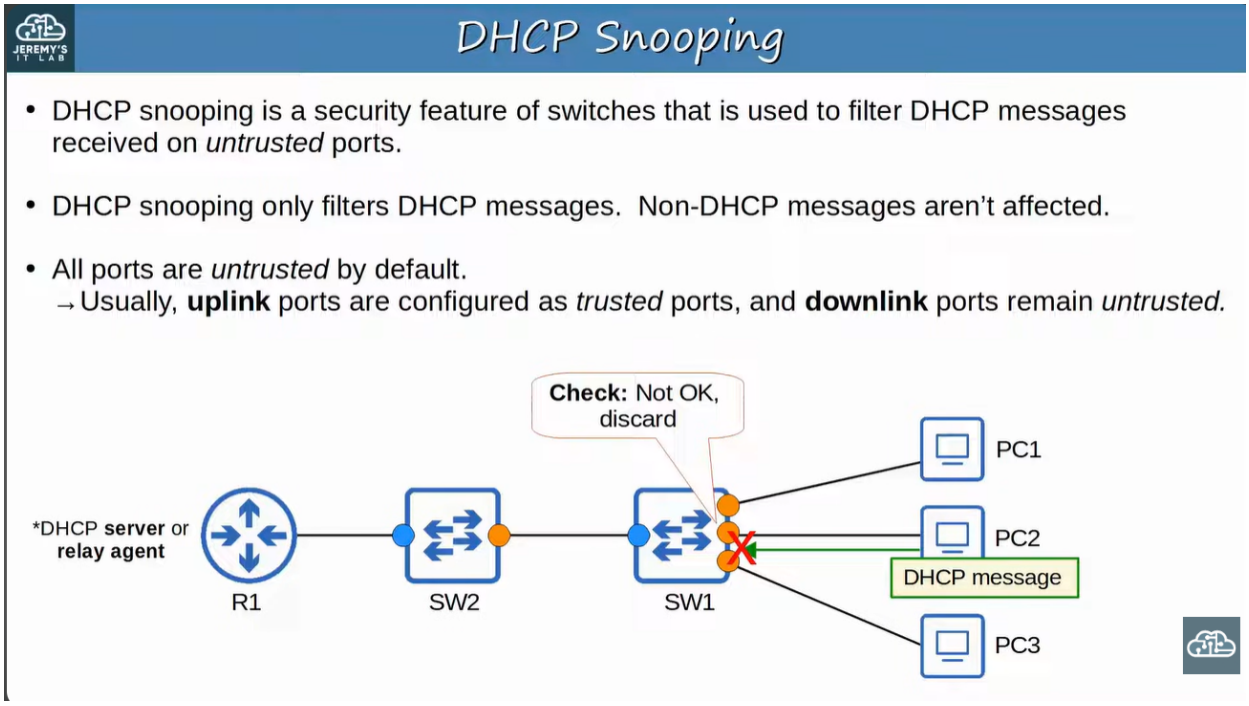


2. Trusted vs Untrusted Ports

- **Trusted port:** DHCP server messages are allowed through normally.
- **Untrusted port:** DHCP server messages received there are inspected and normally discarded.
- Example: R1/legitimate DHCP server → SW2 → SW1 → PCs. The switch ports toward R1 are trusted; PC-facing ports remain untrusted.

Image section:

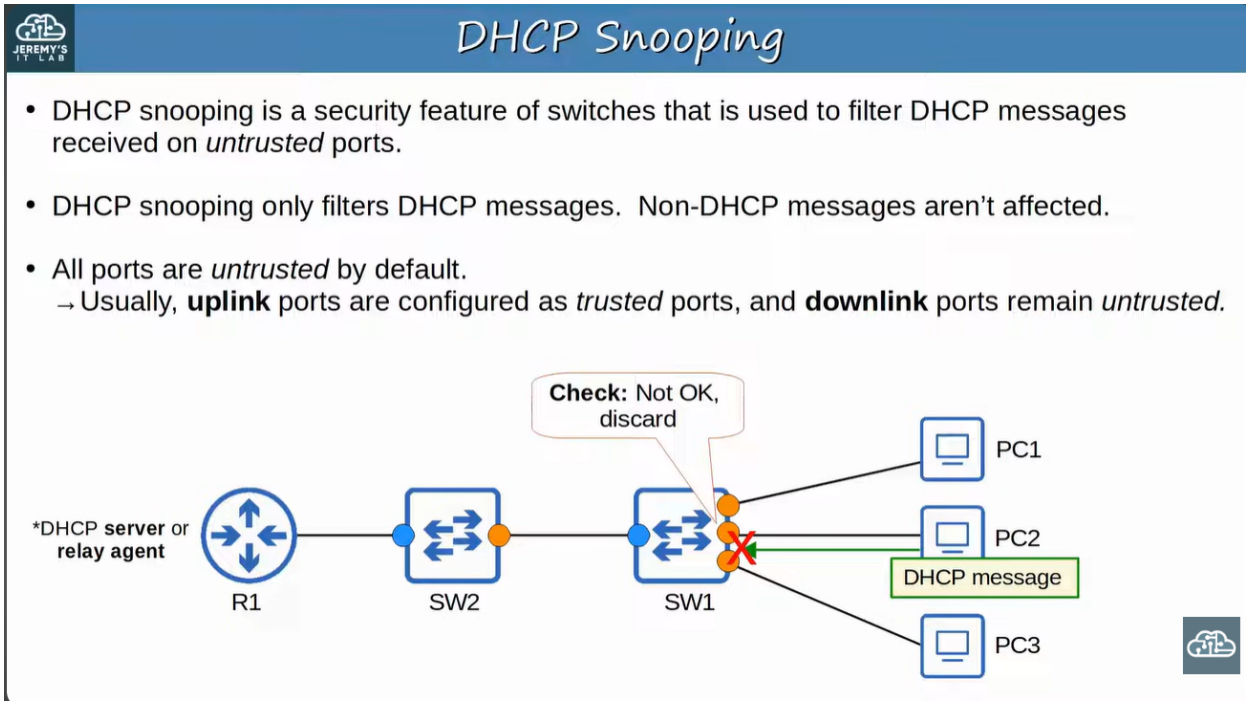




3. DHCP Snooping and Rogue DHCP

- A rogue DHCP server connected to an untrusted client port may try to send **OFFER/ACK** messages.
- DHCP Snooping identifies these as DHCP server messages arriving on an untrusted port and **drops them**.
- This protects clients from receiving incorrect IP configuration from an unauthorized DHCP server.

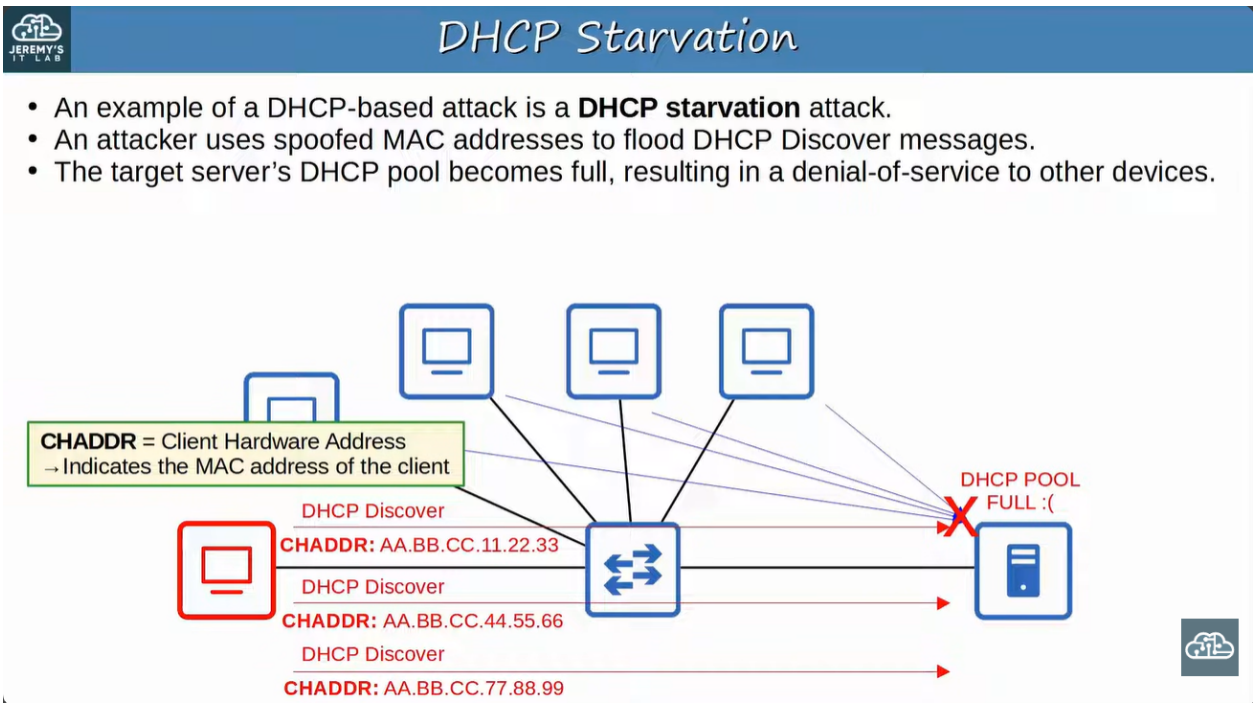
Image section:



4. DHCP Starvation Attack

- The attacker sends many DHCP Discover messages using **spoofed MAC addresses**.
- The legitimate DHCP server keeps allocating addresses until the **DHCP pool becomes full**.
- Legitimate clients can then fail to obtain an IP address: a **Denial-of-Service (DoS)** condition.
- **Defense:** DHCP Snooping rate-limiting can restrict how many DHCP messages enter a port.

Image section:

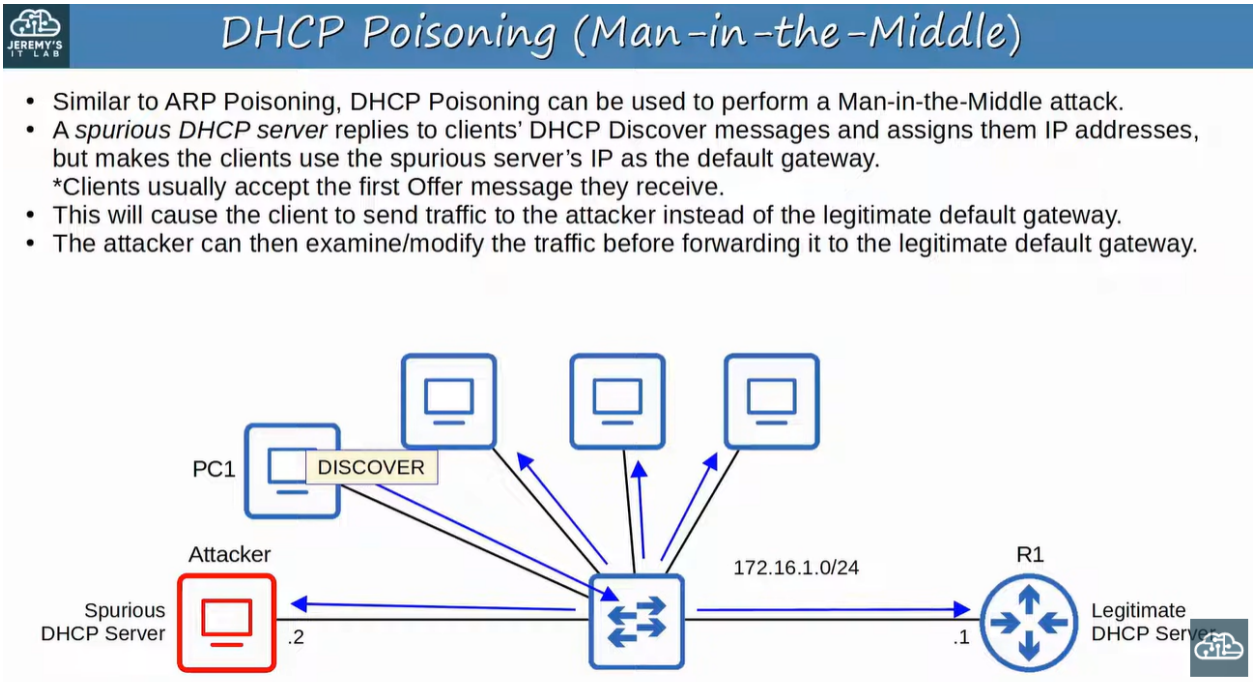


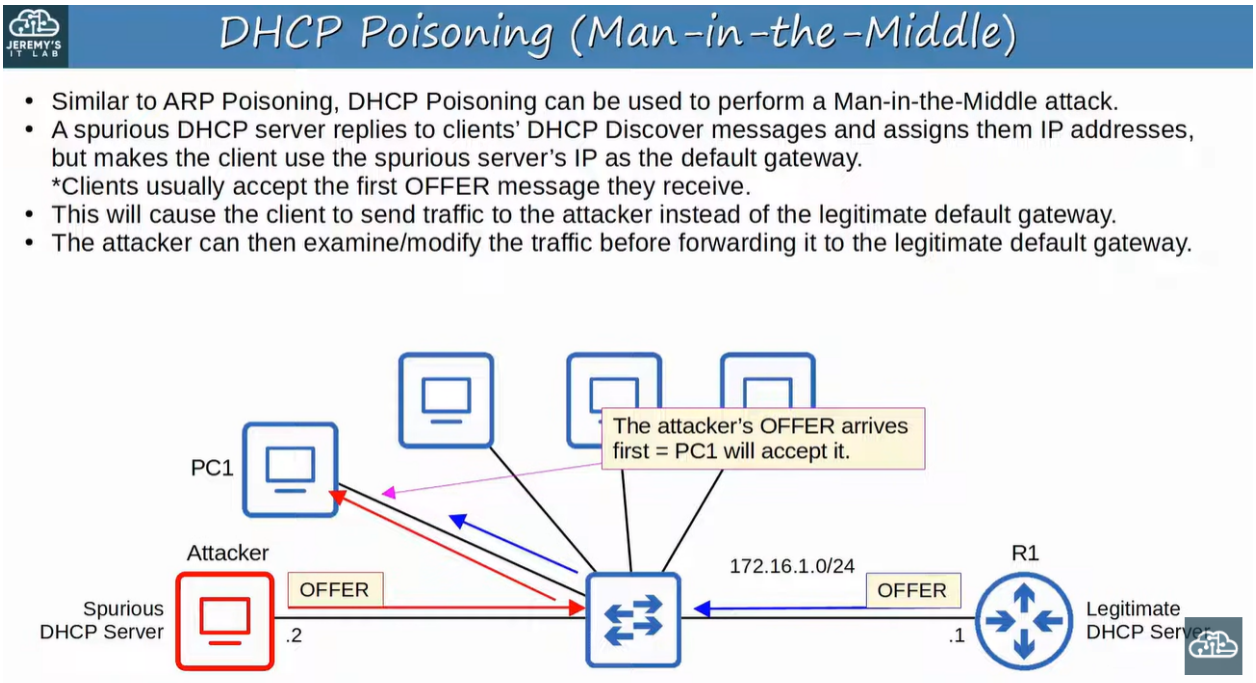
5. DHCP Poisoning / MITM

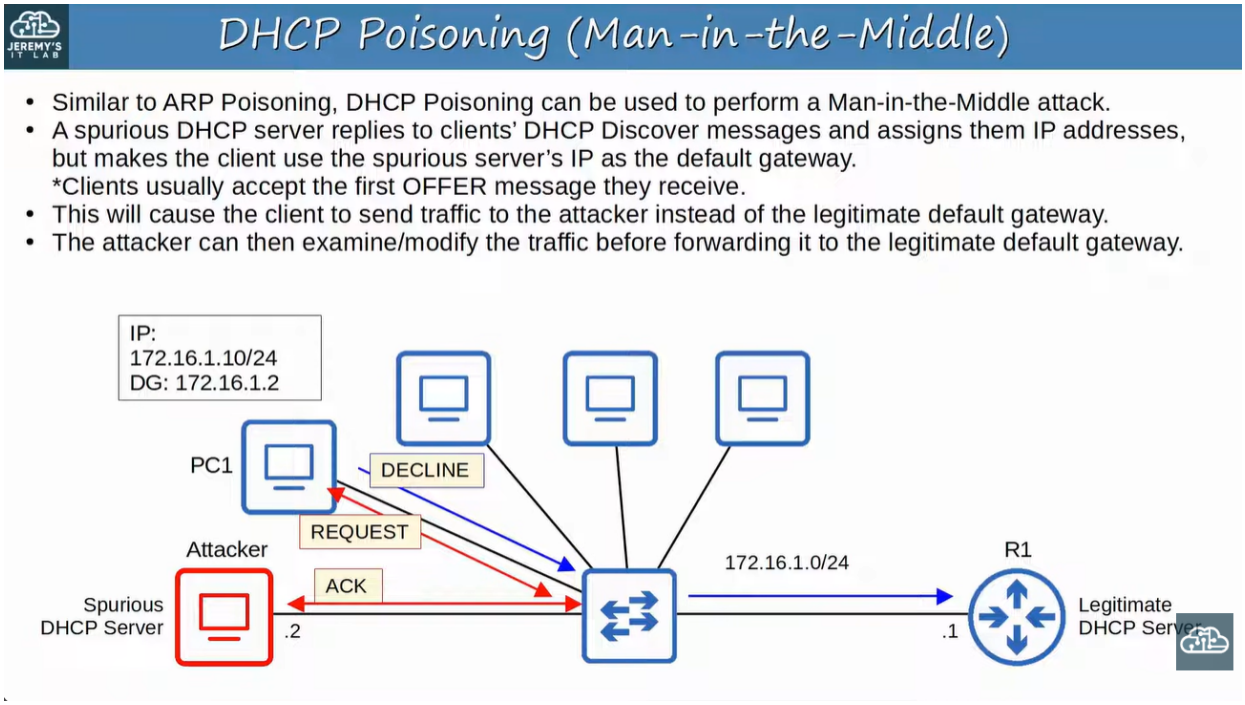
- An attacker runs a **spurious/rogue DHCP server** and replies to clients' DHCP Discover messages.
- The attacker can provide an IP address but advertise the attacker's own IP as the **default gateway**.
- If the client accepts the attacker's OFFER first, traffic can be sent through the attacker.

- The attacker can then inspect or modify traffic before forwarding it toward the legitimate gateway. This is a **Man-in-the-Middle (MITM)** attack.

Image section:



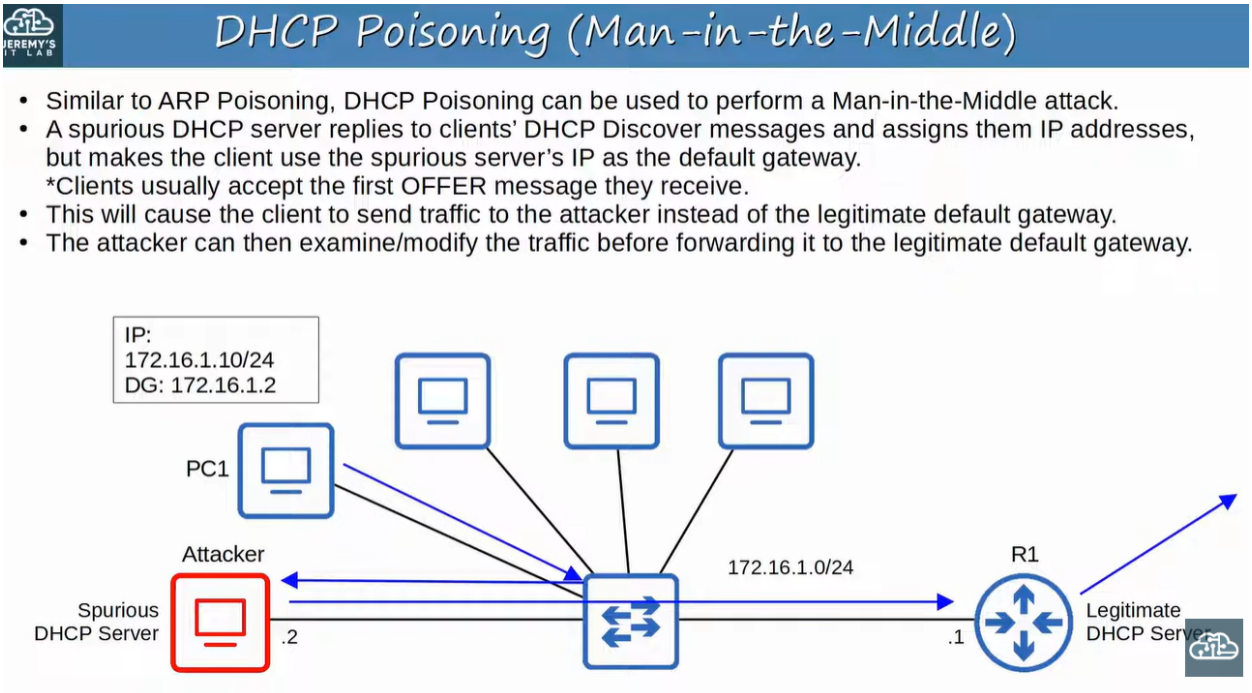




6. DHCP Message Types

- **Client** → **Server:** DISCOVER, REQUEST, RELEASE, DECLINE.
- **Server** → **Client:** OFFER, ACK, NAK.
- **NAK:** server rejects/declines a client's request.
- **RELEASE:** client tells the server it no longer needs its leased IP address.
- **DECLINE:** client rejects the IP address offered by the DHCP server.

Image section:

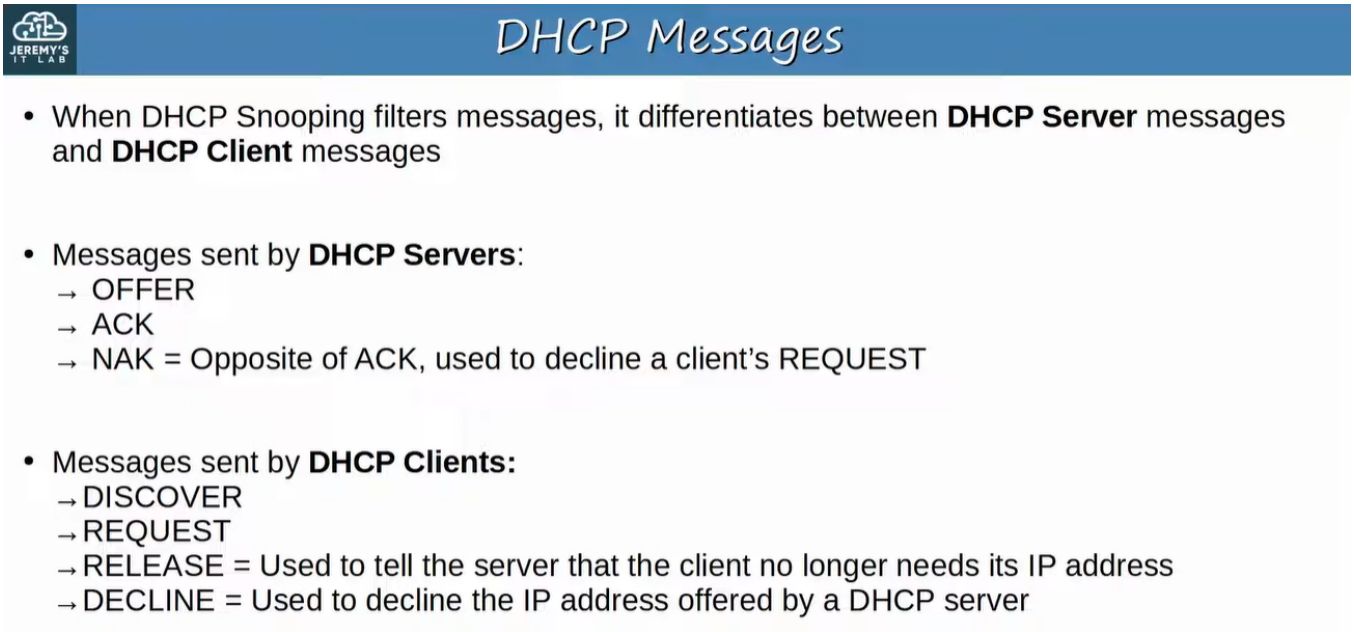


7. DHCP Snooping Operation

- DHCP message on a **trusted port** → forward normally without inspection.
- DHCP message on an **untrusted port** → inspect it.

- If it is a **DHCP Server message** (OFFER/ACK/NAK) → **discard**.
- If it is a **DHCP Client message**, the switch performs additional checks.

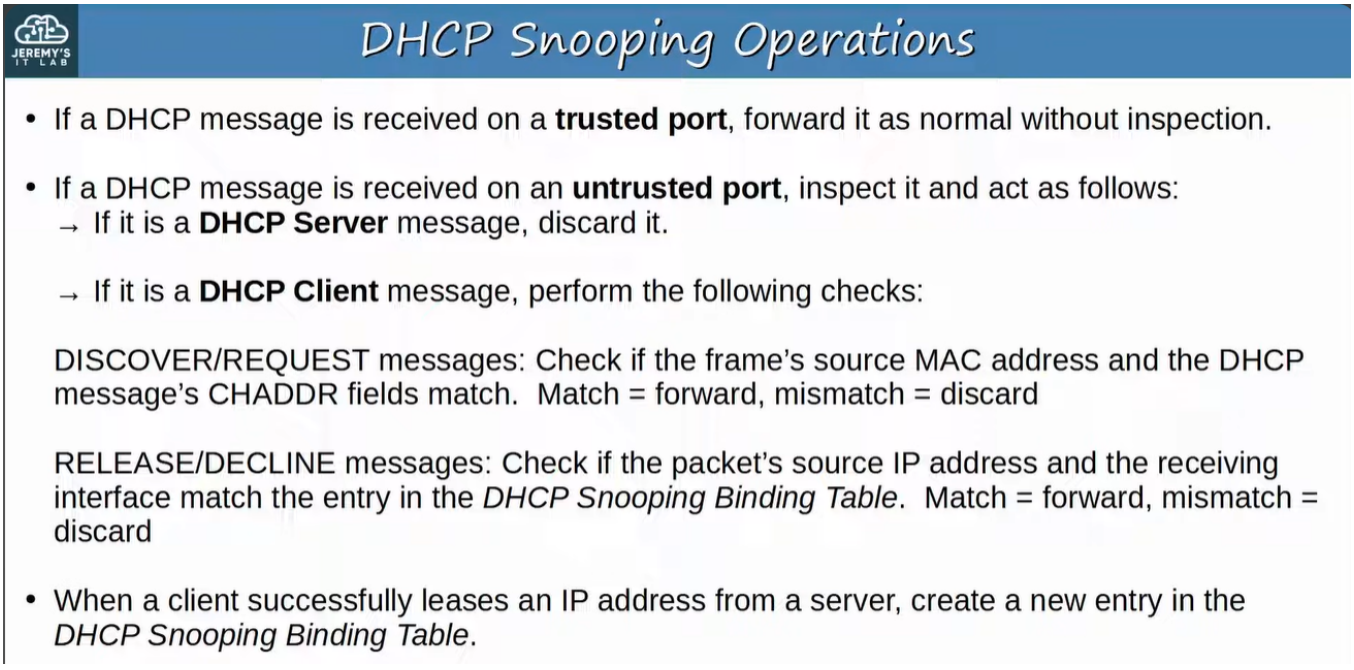
Image section:



8. DHCP Snooping Binding Table

- When a client successfully leases an IP address, the switch creates an entry in the **DHCP Snooping Binding Table**.
- Typical information: **MAC address, IP address, lease time, VLAN, interface**.
- Example command: `show ip dhcp snooping binding`.
- The binding table is also useful for other security features such as **Dynamic ARP Inspection (DAI)** and **IP Source Guard**.

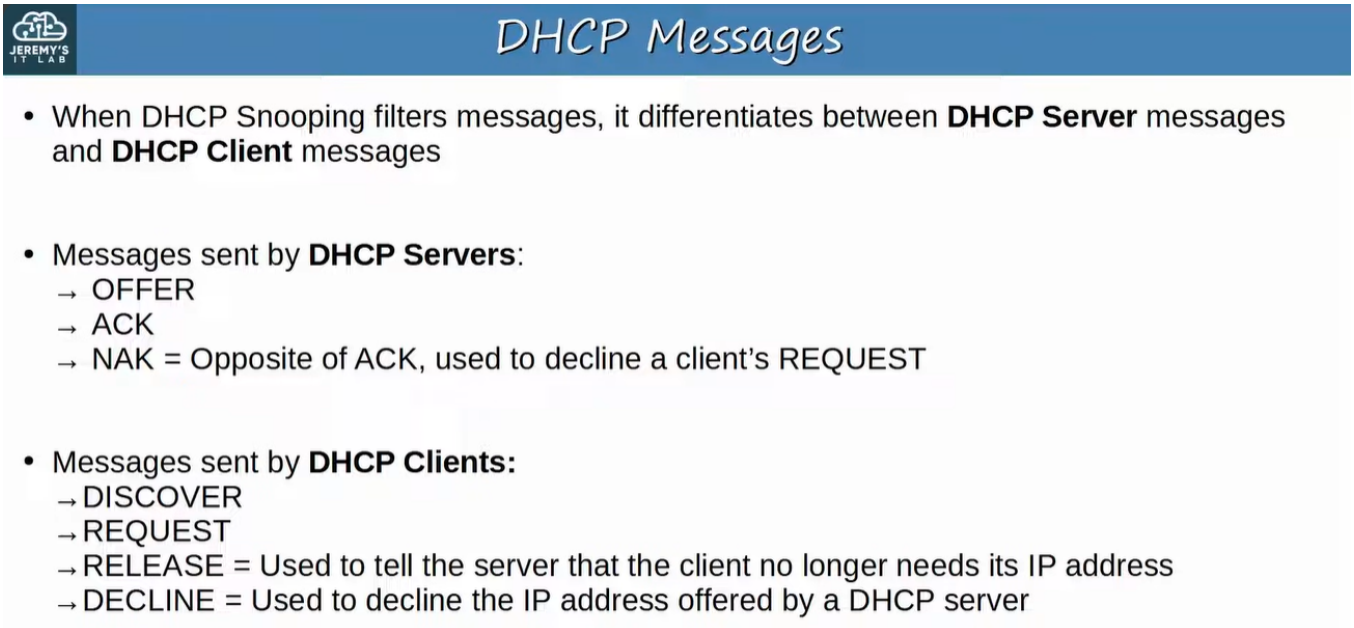
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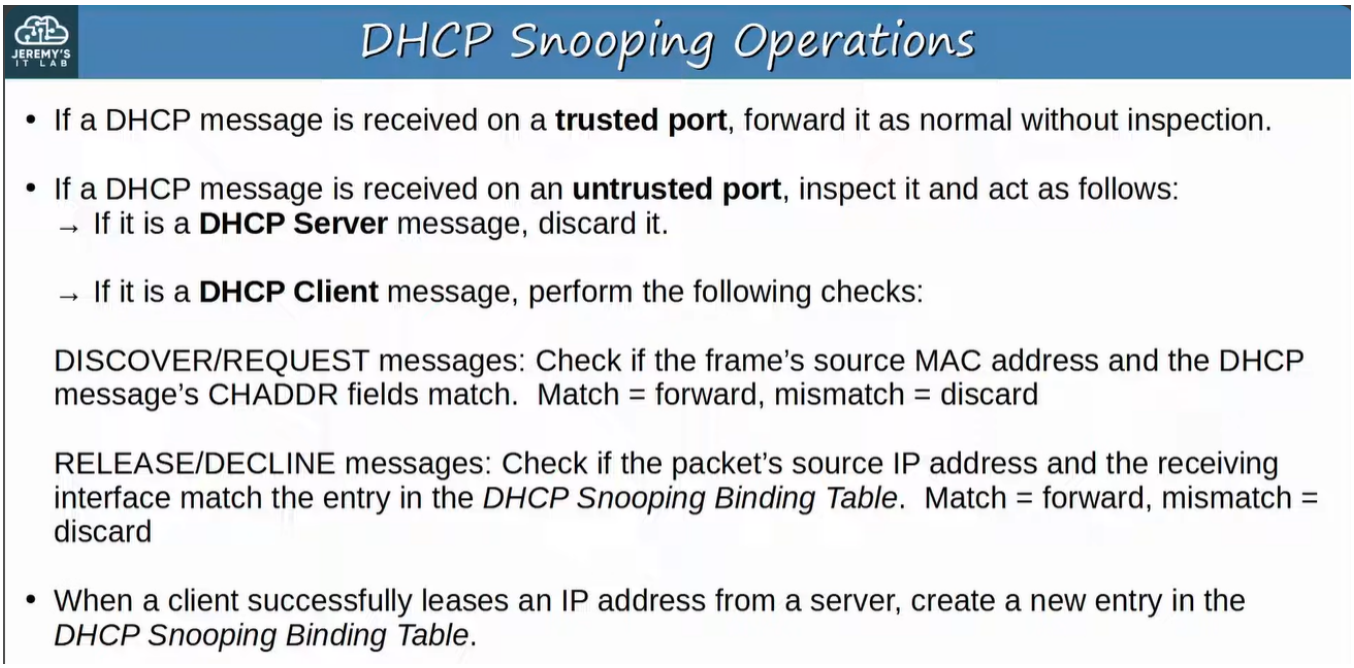


9. Client-Message Checks

- **DISCOVER/REQUEST:** compare the Ethernet frame's source MAC address with the DHCP message's **CHADDR** field. Match → forward; mismatch → discard.
- **RELEASE/DECLINE:** compare the packet's source IP address and receiving interface with the corresponding entry in the DHCP Snooping Binding Table. Match → forward; mismatch → discard.

Image section:

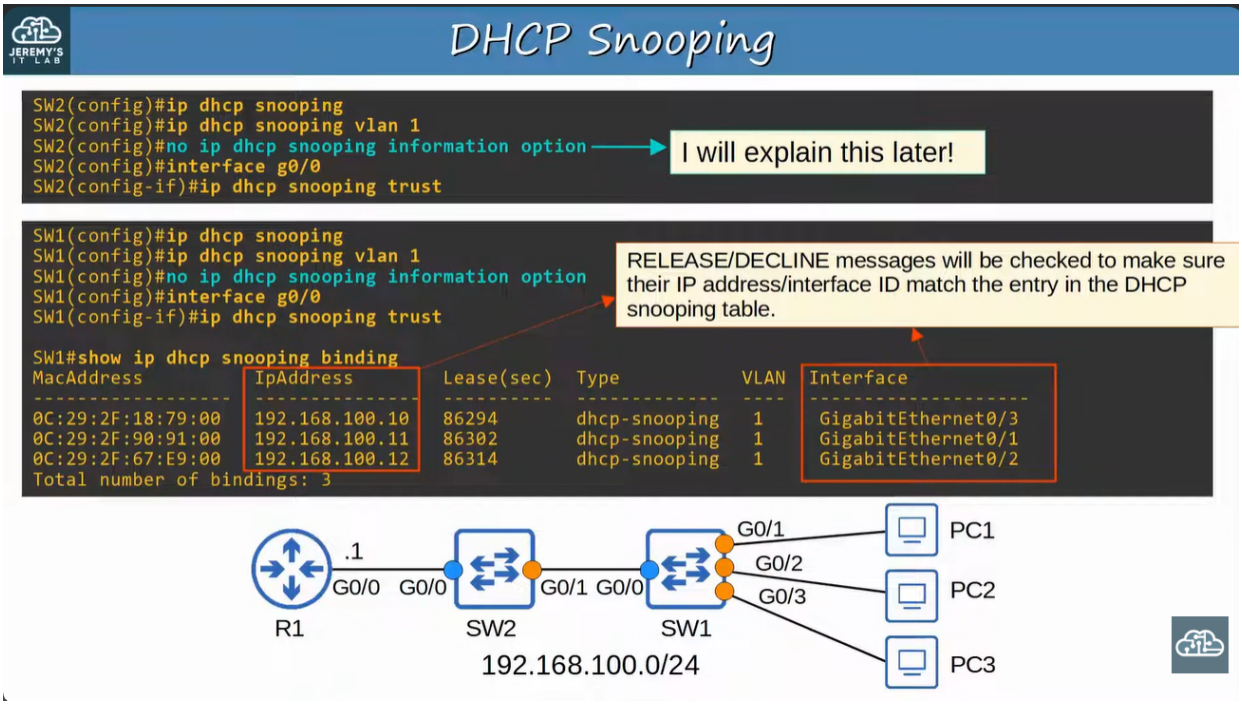


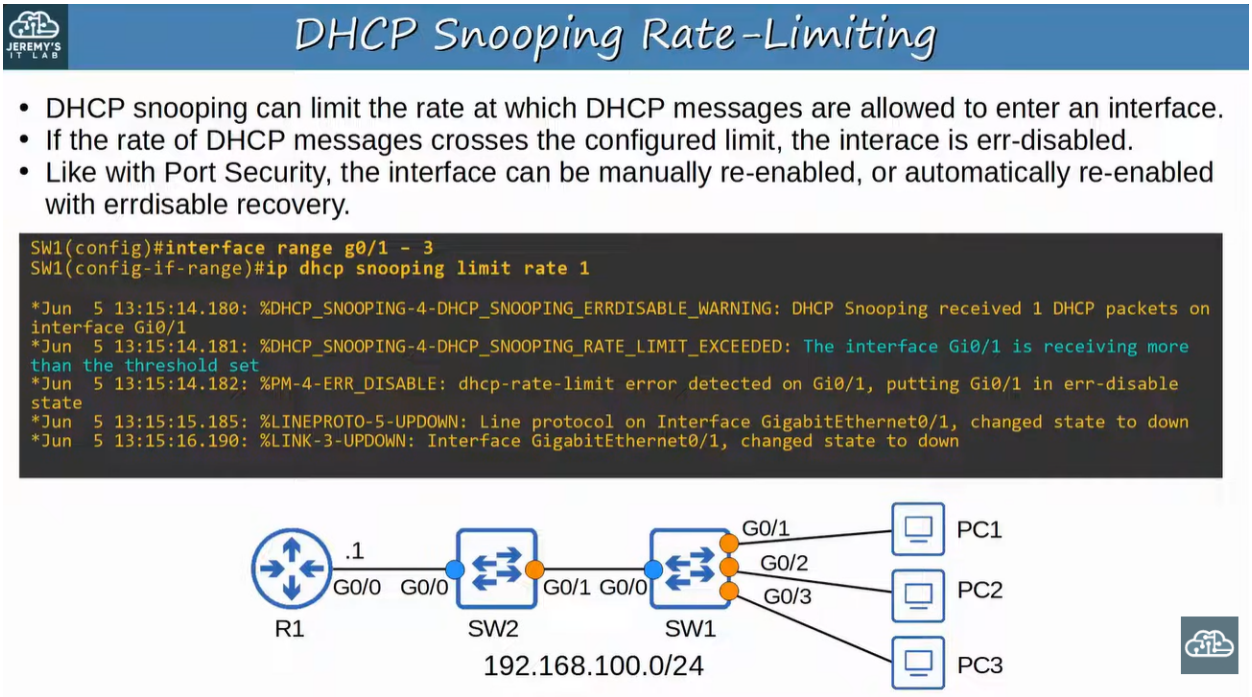


10. DHCP Snooping Rate-Limiting

- Rate-limiting limits how many DHCP messages are allowed to enter an interface per second.
- If the configured rate is exceeded, the interface can be placed into **err-disabled** state.
- This is especially useful against **DHCP starvation/exhaustion attacks**.
- The interface can be manually recovered or automatically recovered using errdisable recovery.

Image section:





11. DHCP Option 82

- **Option 82** = DHCP Relay Agent Information Option.
- It adds information about where the client's DHCP message came from, such as the **relay agent/interface/VLAN**.
- With DHCP Snooping enabled, Cisco switches can add Option 82 to DHCP messages received from clients.
- By default, a Cisco switch can drop DHCP messages containing Option 82 when received on an **untrusted port**.
- To disable insertion of this information option: `no ip dhcp snooping information option`.

Image section:

DHCP Snooping Rate-Limiting

```
SW1(config)#errdisable recovery cause dhcp-rate-limit

SW1#show errdisable recovery
ErrDisable Reason            Timer Status
-----------------            --------------
arp-inspection               Disabled
bpduguard                    Disabled
channel-misconfig (STP)      Disabled
dhcp-rate-limit              Enabled
dtp-flap                     Disabled
gbic-invalid                 Disabled
inline-power                 Disabled
![output omitted due to length]


Timer interval: 300 seconds

Interfaces that will be enabled at the next timeout:

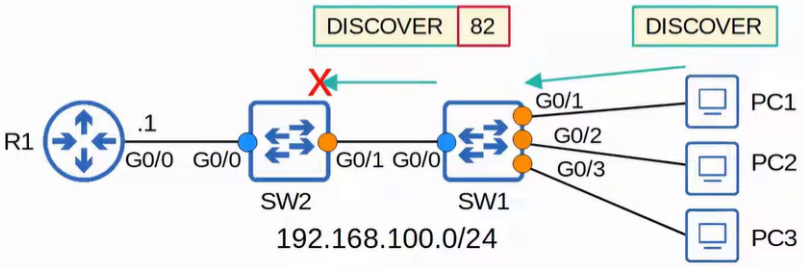
Interface        Errdisable reason        Time left(sec)
---------        -----------------        --------------
Gi0/1             dhcp-rate-limit           293
```

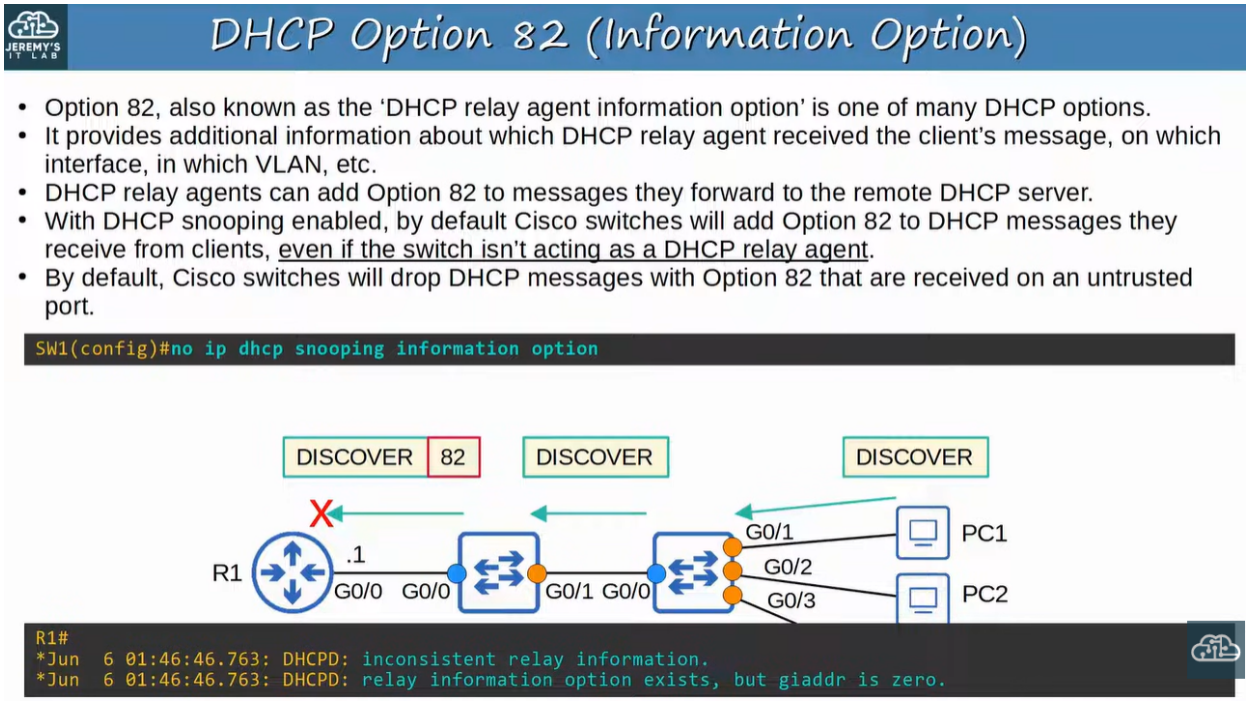
Rate-limiting can be very useful to protect against DHCP exhaustion attacks.

DHCP Option 82 (Information Option)

- Option 82, also known as the 'DHCP relay agent information option' is one of many DHCP options.
- It provides additional information about which DHCP relay agent received the client's message, on which interface, in which VLAN, etc.
- DHCP relay agents can add Option 82 to messages they forward to the remote DHCP server.
- With DHCP snooping enabled, by default Cisco switches will add Option 82 to DHCP messages they receive from clients, even if the switch isn't acting as a DHCP relay agent.
- By default, Cisco switches will drop DHCP messages with Option 82 that are received on an untrusted port.

```
SW2#
*Jun  6 01:36:15.298: %DHCP_SNOOPING-5-DHCP_SNOOPING_NONZERO_GIADDR: DHCP_SNOOPING drop message with non-
zero giaddr or option82 value on untrusted port, message type: DHCPDISCOVER, MAC sa: 0c29.2f67.e900
```

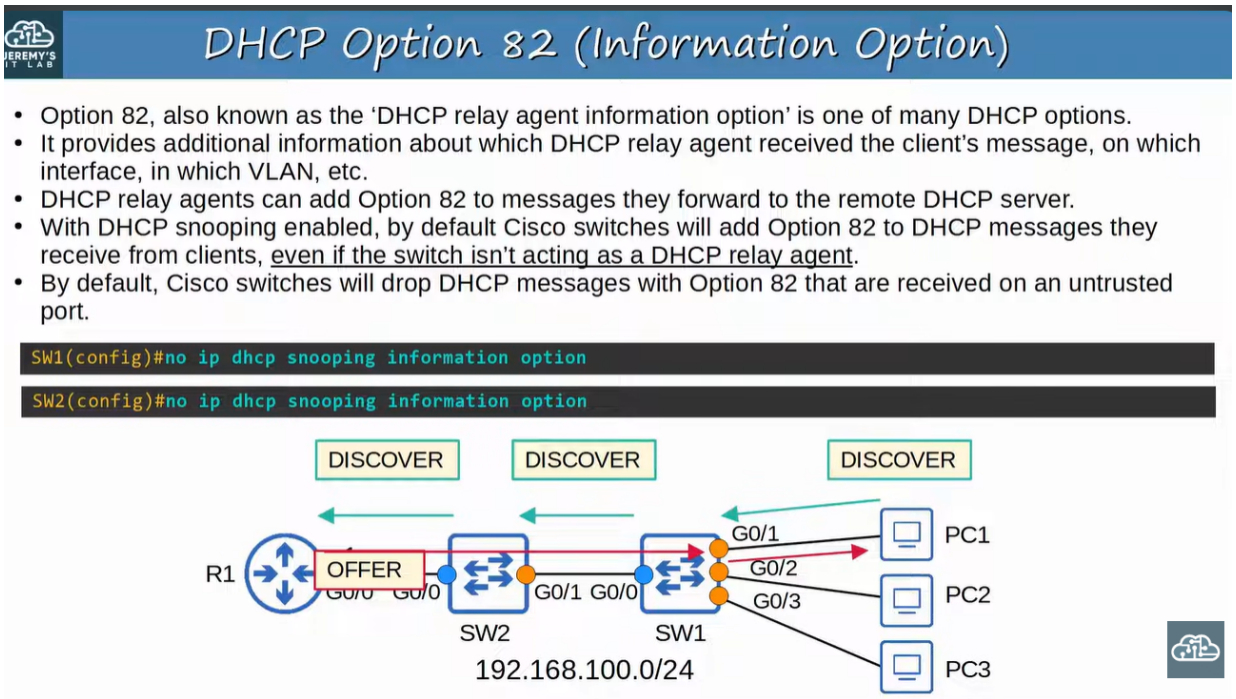




12. Important Cisco Commands

- `ip dhcp snooping` — enable DHCP Snooping globally.
- `ip dhcp snooping vlan 10` — enable it for VLAN 10.
- `ip dhcp snooping trust` — make the interface trusted.
- `ip dhcp snooping limit rate 1` — allow at most 1 DHCP packet per second on that interface.
- `show ip dhcp snooping binding` — display the binding table.
- `errdisable recovery cause dhcp-rate-limit` — enable automatic recovery for this err-disable cause.
- `no ip dhcp snooping information option` — disable DHCP Option 82 insertion.

Image section:



Command Review

```
SW1(config)# ip dhcp snooping

SW1(config)# ip dhcp snooping vlan vlan-number

SW1(config)# errdisable recovery cause dhcp-rate-limit

SW1(config)# no ip dhcp snooping information option

SW1(config-if)# ip dhcp snooping trust

SW1(config-if)# ip dhcp snooping limit rate packets-per-second

SW1# show ip dhcp snooping binding
```

■ 30-Second Revision

- **DHCP Snooping = protection against rogue DHCP servers.**
- All ports are **untrusted by default**.
- Uplink toward legitimate DHCP server/relay = **trusted**.
- DHCP Server messages on untrusted ports = **DROP**.
- Client messages are checked using MAC/CHADDR or the binding table.
- Successful DHCP lease → entry added to **binding table**.
- Rate limiting → helps stop **DHCP starvation**.
- Option 82 → adds relay/interface/VLAN information.
- Key commands: `ip dhcp snooping`, `ip dhcp snooping vlan`, `ip dhcp snooping trust`, `show ip dhcp snooping binding`.